

04_DIGITAL_TWIN_RESULTS

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EVIDENCE PROVENANCE: [SIMULATION BENCHMARK]

Monte Carlo simulation of 50-vehicle fleet operations over 24 hours under cyclic fog at NMDC Bailadila Deposit 5.

1. Fleet Operational Comparison Matrix

Metric	Conventional Baseline	FOG-ORCHESTRATOR 2.0	Operational Impact
Clear Capacity	4,200 t/h	4,200 t/h	Identical baseline capacity
Fog Capacity (<50m)	0 t/h (Total Shutdown)	2,870 t/h	+68.4% throughput retained
Collisions (500 runs)	14 collisions (Blind run)	0 collisions	Zero safety incidents
Switchback Queues	Up to 8 dumpers (Gridlock)	<= 2 dumpers	Eliminates ramp congestion
Fleet Fog Downtime	4.5 hours / shift	0.0 hours	Continuous safe haulage

2. Economic Significance for NMDC Limited

Retaining 2,870 tonnes/hr of iron ore throughput during fog preserves over INR 1.8 to 2.4 Crores per 4-hour fog shift, while entirely eliminating blind driving collision hazards.